

A Study on the Performance of Multi-Channel Communication Protocol for Intelligent Connected Autonomous Driving System

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Abstract—Recently, the development of autonomous driving systems is gradually evolving into a new technology incorporating communication technology for vehicles. Previously, it was merely a warning service due to the limitations of communication technology, but recent developments in communication technology have led to research and development of large-capacity data transmission or vehicle guidance technology utilizing low latency. In this paper, we propose a multi-hop transmission protocol technology using multiple channels to improve the reliability and delay time of communication networks. Through the simulation, the proposed multi-channel algorithm and route routing metric were found to significantly improve performance as the number of data transfer hops increases.

Keywords—CAV, Connected, Multi-channel, V2X, ITS, Mesh

I. INTRODUCTION

Recently, with the rapid development of autonomous driving technology, various advanced driver assistance functions have been widely distributed to commercial vehicles. Overseas, companies such as Google Waymo, GM Cruise, and Tesla Model 3 are releasing commercial vehicles of level 3 autonomous driving system. As of the end of 2020, more than 130 vehicles have acquired a temporary autonomous driving license in Korea. Until now, most autonomous driving technologies have developed technologies that rely on expensive sensors to perform object recognition, planning, and driving control [1]. For the development of autonomous driving technology, intelligent connected vehicle (ICV) using vehicular communication technology is drawing attention. In the beginning of 2010, study on the vehicle communication technology was focused on basic safety services, transmission speed and interference for safe driving [2,3]. However, with the standardization of 5G-NR V2X technology, the data transmission capacity and latency requirements of autonomous vehicles have improved, and many use cases of cooperative autonomous driving systems using V2X technology have been introduced.

The challenge to be solved in vehicular wireless communication technology is to successfully transmit data without data collision. When transmitting and receiving data through multi-hop between vehicles, it is necessary to solve the problem of optimizing the maximum network capacity and latency by maximizing the given channel and network interface. The multi-interface multi-channel (MIMC) technology has been studied as an alternative to overcome this problem. In [4], some common dynamic channel allocation algorithms are analyzed and the necessity and effectiveness of

channel allocation in WMN are explained. Considering the issue of static channel allocation for a multi-radio, multi-channel 802.11 wireless mesh network, the author in [5] have presented four metrics for the mesh channel allocation. Traffic must be balanced across all available channels to achieve the maximum possible load on a wireless communications network. In [6], the analysis relationship between throughput, node utilization, number of transmissions, and channel utilization was derived by considering interference constraints. Most of the research to date has been on WLAN-based channel allocation method and network optimization method. In order to be used as a network technology for autonomous driving, research on channel allocation for load balancing and low latency is required based on vehicle communication technology. In this paper, we introduce a channel allocation method and simulation results based on low latency MIMC.

In this paper, we analyze the latency from the source to the destination node by employing the proposed multi-channel allocation algorithm in a street environment. The system architecture for MIMC network, algorithm and numerical analysis are described in Section 2, and the performance of the multi-channel allocation scheme is simulated in Section 3. Finally, some concluding remarks are presented in Section 4.

II. SYSTEM MODELING

A. System Architecture

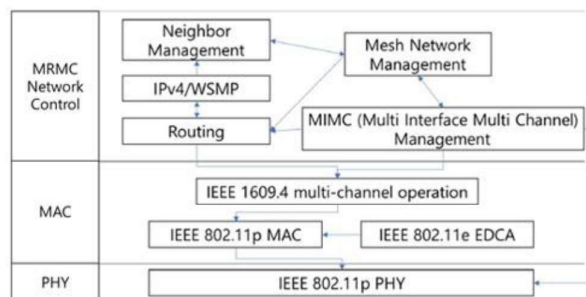


Fig. 1. V2X unit system architecture for MRMC network

The communication protocol stack for low latency mesh network based on MIMC scheme is shown in Fig.1. It is

composed of IEEE 802.11p for physical layer, IEEE 802.11e for EDCA, IEEE 1609.4 for multi-channel operation, WSMP for safety message transmission, and management blocks for MIMC mesh network. The developed mesh network module establishes and manages a mesh network through interaction with neighbor nodes, forms an optimal communication path through multi-hop through the operation of the routing function. The developed module efficiently manages 2 interfaces and 7 channels through the MIMC function.

B. Multichannel Allocation Algorithm

The flow of message operations performed by communication nodes in the service channel and the common channel interval for MIMC is shown in Fig.2. When the current channel section is a service channel interval, the communication device transmits/receives service data or measures the degree of interference for available service channels. In the case of a common channel interval, control messages such as RANN, Hello, PREQ, and PREP can be transmitted and received. If a communication node has selected a service channel, a Hello message is broadcasted to the surroundings in a Hello(S) type.

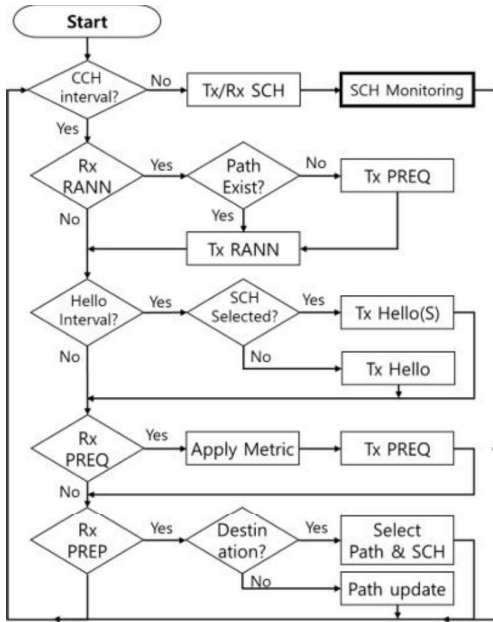


Fig. 2. Block diagram for channel allocation at V2X node

C. Routing Metric of Expected Transmission Time

In a mesh network, when configuring a multi-hop communication path, an optimal path is selected using a routing metric. In this paper, a metric that minimizes the communication delay time is used. Each node computes the probability of successive packet reception (p_r) and the probability of successive packet transmission (p_f). Then the probability of packets not being sent or received successively as

$$p = 1 - (p_f + p_r + p_f \times p_r) \quad (1)$$

Using P , the probability of packet forwarding after k -th attempt can be obtained as

$$s(k) = p^{k-1} \times (1 - p) \quad (2)$$

Therefore, the expected value of the number of retransmissions and time at the node is calculated as follows.

$$ETT = \frac{1}{B} \sum_{k=1}^{\infty} k \times s(k) = \frac{1}{B} \times \frac{1}{1-p} \quad (3)$$

where B is the bandwidth of the transmission signal.

ETT is used to select the path with the least delay by estimating the delay time of the link. Actually, in this study, the probability, p is calculated by measuring the RSS value. That is, assuming that the RSS value follows a Gaussian distribution, the probability p is calculated by measuring the ratio of RSS that exceeds the reception threshold of the node.

III. SIMULATION RESULTS

This chapter introduces the simulation results of transmitting information from the node where the event occurred to the destination using the MIMC-based mesh technology. The communication node provides two RF interfaces and considers one control channel and up to six service channels, and detailed parameter information is shown in Table 1.

TABLE I. SIMULATION PARAMETERS

Parameters	Value	Description
1 hop distance	60m	1 hop distance between nodes
Tx Power	25dBm	EIRP
Channels	CCH, SCH1~6	CCH for control message SCH for data service
Frequency	5.9GHz	ITS band
Bandwidth	10MHz	per channel
Channel model	Log distance	Propagation loss model
Multichannel	Alternative mode	50ms period
Data rate	27Mbps	Data rate for simulation

It is assumed that 16 connected vehicles are arranged at intervals of 20 to 70 meters in a typical crossroad road environment. A packet of 1300 bytes is generated in the event vehicle every 1ms. To evaluate the multi-channel allocation algorithm, we compare the performance of Multi-Channel, Double-Channel, and Single-Channel scenarios. Multi-Channel is a case of using up to 6 SCHs, Double-Channel is a case of using two fixed channels, and Single-Channel is a case of using a single channel.

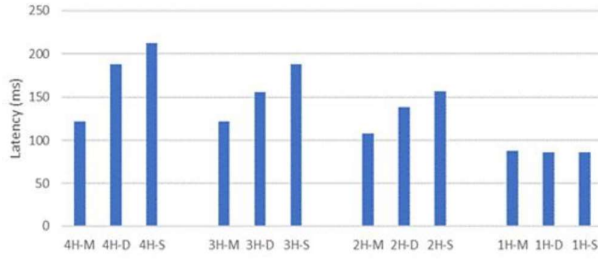


Fig. 3. The mean delay comparison for the different hop and channel numbers

The performance comparison in delay time according to the number of data transfer hops and the number of channels is shown in Fig.3. In 2~4 Hop conditions, Multi-Channel shows the highest performance. In particular, in the Multi-Channel scheme, as the number of hops increases, the increase in link delay time is significantly less than that of other schemes. This is because in a multi-hop environment, multiple channels are efficiently allocated and traffic is distributed to multiple channels, reducing retransmissions due to wireless communication collisions.

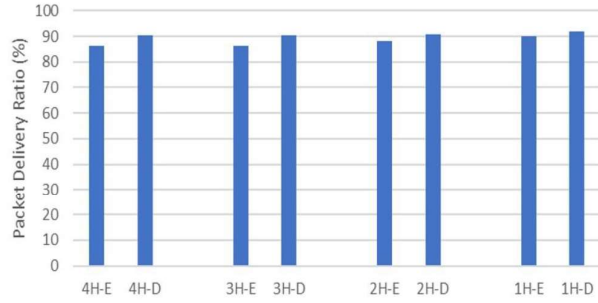


Fig. 4. PDR performance for different routing metric (ETT vs. DV)

The performance comparison of PDR according to ETT and DV (Distance Vector) metrics is shown in Fig.4. The performance of DV shows about 2~3% higher than that of CBETT because DV has less overhead, and once the path is set based on the minimum hop, it does not modify the path again.

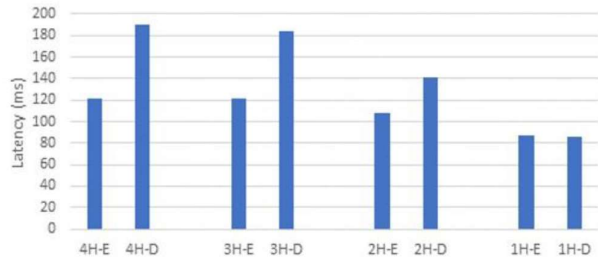


Fig. 5. Latency performance for different routing metric (ETT vs. DV)

The comparison of link latency between ETT and DV metrics is shown in Fig.5. In the 4-hop scenario, the link delay

time of ETT is about 120ms, and the average of 70ms or more packets was delivered faster than the 190ms delay time of DV. In addition, ETT showed lower link latency performance than DV in 3Hop and 2Hop scenarios.

IV. CONCLUSIONS

In this paper, we proposed a multi-channel allocation method and a routing algorithm to minimize the data transmission delay time of the connected vehicle. In order to guarantee the reliability and delay time of data transmission, a mesh network technology based on MIMC was proposed and verified through simulation. Through simulation, it was found that the performance of the proposed algorithm significantly improved as the number of multi hops increased or the number of available service channels increased.

ACKNOWLEDGMENT

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